

EX. NO: 2

MONOALPHABETIC CIPHER

Aim:

To write a C program to implement the Monoalphabetic Cipher, which encrypts a plaintext using a fixed substitution key derived from a permutation of the alphabet.

Algorithm:

1. Start the program.
2. Define a substitution key array containing a permutation of the twenty-six letters of the alphabet.
3. Read the plaintext from the user.
4. Convert each character of the plaintext to uppercase.
5. For every alphabetic character, replace it with the character present at the corresponding position in the substitution key.
6. Leave non-alphabetic characters unchanged.
7. Display the encrypted text.
8. Stop the program.

Program:

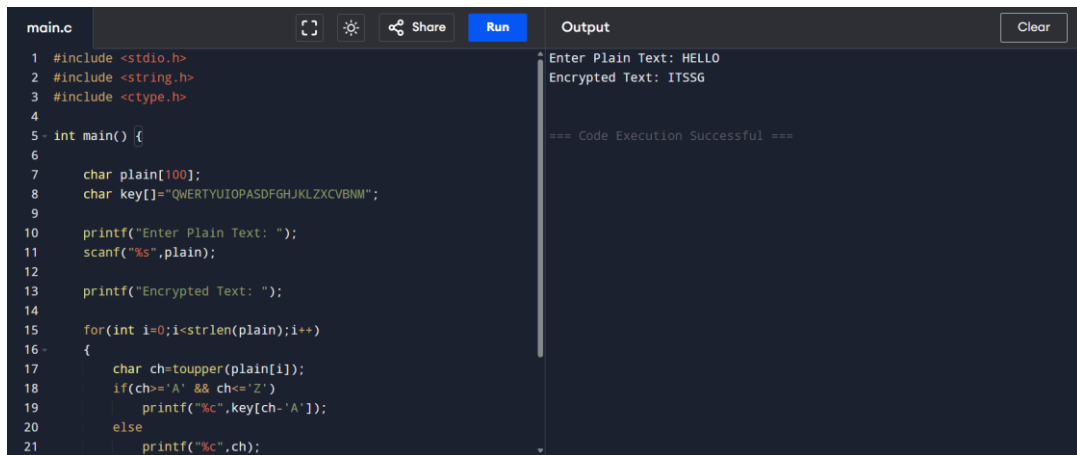
```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char plain[100];
    char key[] = "QWERTYUIOPASDFGHJKLZXCVBNM";

    printf("Enter Plain Text: ");
    scanf("%s", plain);

    printf("Encrypted Text: ");
    for (int i = 0; i < strlen(plain); i++) {
        char ch = toupper(plain[i]);
        if (ch >= 'A' && ch <= 'Z')
            printf("%c", key[ch - 'A']);
        else
            printf("%c", ch);
    }
    printf("\n");
    return 0;
}
```

Output:



```
main.c  [Icons]  Run  Output  Clear

1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int main() {
6
7     char plain[100];
8     char key[]="QWERTYUIOPASDFGHJKLZXCVBNM";
9
10    printf("Enter Plain Text: ");
11    scanf("%s",plain);
12
13    printf("Encrypted Text: ");
14
15    for(int i=0;i<strlen(plain);i++)
16    {
17        char ch=toupper(plain[i]);
18        if(ch>='A' && ch<='Z')
19            printf("%c",key[ch-'A']);
20        else
21            printf("%c",ch);
22    }
23 }
```

Enter Plain Text: HELLO
Encrypted Text: ITSSG

=== Code Execution Successful ===

Result:

Thus a C program to implement the Monoalphabetic Cipher using a substitution key was written, executed and the output was verified successfully.